

SOAP2REST Bridge

Document 06 - Implementation Plan

Project Goal

Build an AI-powered SOAP/XML to REST/JSON Bridge using Spring Boot 3.x, Java 21, Maven, React, TypeScript and Tailwind CSS. The platform should parse WSDL files, generate AI-powered field mappings, create REST APIs dynamically and act as a runtime proxy between REST and SOAP services.

Phase 1: Project Setup & Environment Configuration

Objectives: • Create Maven multi-module project structure • Configure Spring Boot 3.x and Java 21 • Initialize React + Vite + TypeScript frontend • Configure Git repository • Configure environment variables • Setup Swagger/OpenAPI Deliverables: ✓ Backend project created ✓ Frontend project created ✓ Git repository configured ✓ OpenAPI working

Phase 2: Database Design & JPA Entities

Objectives: • Create database schema • Implement JPA entities • Configure H2/SQLite • Create repositories • Configure entity relationships Deliverables: ✓ All tables created ✓ JPA repositories implemented ✓ Database connectivity verified

Phase 3: Authentication & Authorization

Objectives: • Implement JWT authentication • Create login/register APIs • Configure Spring Security • Create API Key management module • Implement RBAC permissions Deliverables: ✓ JWT security enabled ✓ API Keys generated securely ✓ Role-based access working

Phase 4: WSDL Upload & Parsing Engine

Objectives: • Build WSDL upload API • Parse XML using DOM/SAX parsers • Extract SOAP operations • Extract request/response schemas • Store metadata in database Deliverables: ✓ WSDL files parsed successfully ✓ SOAP operations extracted ✓ Metadata persisted

Phase 5: AI Mapping Agent

Objectives: • Integrate Gemini API/Ollama • Generate semantic field mappings • Convert SOAP naming conventions • Store mapping results • Allow manual mapping approval Deliverables: ✓ AI mappings generated ✓ Mapping review screen operational ✓ Confidence scores stored

Phase 6: Dynamic REST Route Generation

Objectives: • Generate REST endpoint definitions • Create route registry • Build dynamic endpoint engine • Generate Swagger documentation Deliverables: ✓ REST endpoints generated ✓ OpenAPI documentation available ✓ Route registry operational

Phase 7: Runtime Proxy Engine

Objectives: • Build JSON → SOAP transformer • Build SOAP → JSON transformer • Implement SOAP client dispatcher • Handle request/response conversion • Implement exception handling Deliverables: ✓ End-to-end proxy execution working ✓ XML conversion validated ✓ Error handling implemented

Phase 8: Frontend Dashboard Development

Objectives: • Create dashboard UI • Build WSDL upload screen • Build mapping console • Build API management module • Build API testing sandbox Deliverables: ✓ Dashboard completed ✓ API console functional ✓ Responsive UI available

Phase 9: Analytics & Monitoring

Objectives: • Capture transaction logs • Monitor latency • Track API usage • Generate analytics charts • Build reporting dashboard Deliverables: ✓ Analytics dashboard completed ✓ Performance metrics tracked ✓ Request logs visible

Phase 10: Mock SOAP Service

Objectives: • Create sample SOAP backend • Implement test operations • Simulate enterprise SOAP responses • Enable local testing Deliverables: ✓ Mock SOAP service operational ✓ End-to-end testing enabled

Phase 11: Testing & Quality Assurance

Objectives: • Unit testing • Integration testing • Security testing • Performance testing • User acceptance testing Deliverables: ✓ Test cases passed ✓ Security validated ✓ Performance benchmarks achieved

Phase 12: Deployment & Production Release

Objectives: • Containerize application using Docker • Configure CI/CD pipeline • Deploy backend and frontend • Configure monitoring tools • Production validation Deliverables: ✓ Production deployment completed ✓ Monitoring active ✓ Documentation finalized

Project Timeline

Week 1 : Setup + Database Week 2 : Security + WSDL Parser Week 3 : AI Mapping Engine Week 4 : Dynamic Route Generation Week 5 : Runtime Proxy Engine Week 6 : Frontend Dashboard Week 7 : Analytics + Monitoring Week 8 : Testing + Deployment

Done Criteria

✓ User uploads WSDL successfully ✓ SOAP operations extracted automatically ✓ AI generates field mappings ✓ REST APIs generated dynamically ✓ JSON ↔ SOAP conversion working ✓ Authentication secured with JWT ✓ API Keys functional ✓ Analytics dashboard operational ✓ Mock SOAP service available ✓ OpenAPI documentation generated ✓ Production deployment successful